CCNA 200-301 Enterprise Mega Lab

REVISION 3

Comprehensive Network Architecture, Physical Interconnection Matrix, Addressing Blueprint, 4-Phase Implementation & Verification Guide, and Complete Cisco IOS Startup Configurations

Executive Architectural Scope

This engineering workbook provides an end-to-end, production-grade enterprise network topology designed for Cisco Packet Tracer and validated on Packet Tracer 9.0.1. It integrates every core technical domain of the Cisco Certified Network Associate (CCNA 200-301) blueprint across a redundant Headquarters (HQ) Collapsed Core/Distribution campus, a dual-homed ISP WAN edge, and a remote Branch office.

CCNA Exam Domain	Protocols & Technologies Implemented	Target Topology Tier & Devices
1. Layer 2 Switching & Redundancy	VLANs (Data, Voice, Server, Mgmt, Native 999, Blackhole 888), 802.1Q Trunks with VLAN Pruning, DTP Disabled, LACP EtherChannel (`Po1`–`Po5`), Rapid PVST+ Root Alignment, PortFast & BPDU Guard, HSRP v2	`HQ-DSW1`, `HQ-DSW2` (Catalyst 3650-24PS) `HQ-ASW1`, `HQ-ASW2`, `BR-ASW1` (Catalyst 2960-24TT)
2. Layer 3 Routing & WAN Resilience	Multi-Area OSPFv2 (Area 0 Backbone & Area 1 Branch), Deterministic Router-IDs, Point-to-Point & Broadcast OSPF Network Types, Passive Interfaces, Cost Tuning, Router-on-a-Stick (RoAS), Primary & Floating Static Default Routes (`AD 1` / `AD 5`)	`ISP-R1`, `HQ-R1` (ISR 4331) `BR-R1` (ISR 2911) `HQ-DSW1`, `HQ-DSW2` (L3 SVIs & Routed Ports)
3. IP Infrastructure Services	Centralized Server DHCP & Local Cisco IOS DHCP Pools, Voice VLAN Option 150, Redundant DHCP Relay (`ip helper-address`), Static Inside NAT, PAT Overload via Public NAT Pool (failover-safe), Stratum 2 NTP Master & Clients	`HQ-R1` (NAT/PAT & NTP Master) `HQ-DSW1`, `HQ-DSW2` (DHCP Relay) `BR-R1` (IOS DHCP Server), `HQ-SRV1`
4. Network Security & Hardening	Sticky MAC Port Security (`restrict` & `shutdown`), Extended Named Inter-VLAN ACLs, Standard Named VTY ACLs, SSHv2 (2048-bit RSA), Blackhole Parking	All 8 Routers & Switches (`ISP-R1`, `HQ-R1`, `BR-R1`, `HQ-DSW1`, `HQ-DSW2`, `HQ-ASW1`, `HQ-ASW2`, `BR-ASW1`)

Workbook Reference Summary

Workbook Specification	Addressing Architecture	Default Credentials	Target Environment
CCNA 200-301 Blueprint 4-Phase Lab & Full CLI	Campus/Branch: `10.0.0.0/8` Transit/Lo0: `172.16.0.0/16` Public WAN: `203.0.113.0/24` HQ Public NAT: `203.0.113.64/29`	Enable: `CiscoCCNA#Enable1` SSH User: `admin` SSH Pass: `CiscoCCNA#Admin1`	Cisco Packet Tracer 9.0.1 (validated) ISR 4331 / 2911 & Catalyst 3650 / 2960

Revision 3 Change Log

Area	Change	Reason
HQ-R1 Branch link hardware	Slot 2 module is `NIM-ES2-4` (not `NIM-2GE`); `G0/2/0` is a switchport in VLAN 1 and the routed address `172.16.1.1/30` moved to `interface Vlan1`; unused `G0/2/1`–`G0/2/3` shut	Packet Tracer's ISR4331 has no `NIM-2GE`; `NIM-ES2-4` ports are Layer 2 and reject `ip address`, so OSPF to BR-R1 never formed
OSPF network type	`ip ospf network point-to-point` removed from `BR-R1 G0/0` and from `Vlan100` on both DSWs; these links run as broadcast (DR/BDR)	Packet Tracer ignores point-to-point on VLAN interfaces; the resulting type mismatch with BR-R1 stopped Branch routes installing
Port security (HQ-ASW1 F0/1)	`maximum 2` raised to `maximum 5`	The phone's MAC is learned in both the data and voice VLANs, which filled both slots and blocked the phone with a violation
DHCP Snooping & DAI removed	All `ip dhcp snooping` and `ip arp inspection` commands removed from `HQ-ASW1` and `HQ-ASW2` (global, VLAN, trust and rate-limit lines)	With snooping on, Packet Tracer 9.0.1 dropped the relayed DHCP replies for HQ PCs, over EtherChannel and plain trunks alike, even after a reload. DAI depends on the snooping binding table, so it was removed with it
Validation	Lab validated end to end on Packet Tracer 9.0.1; restart tip added to Phase 3	A stale DHCP state in Packet Tracer was cleared only by restarting the application

Revision 2 Change Log

Area	Change	Reason
NAT / PAT & WAN failover	PAT moved from interface `G0/0/0` to pool `HQ_PAT` (`203.0.113.65`); HQ-SRV1 Static NAT moved from `203.0.113.130` to `203.0.113.66`; `ISP-R1` routes `203.0.113.64/29` to HQ over the primary (AD 1) and floating (AD 5) links	PAT failed on the Serial backup because translations were tied to `G0/0/0`; `.130` sat inside the ISP DMZ `/25`, so ISP-WEB treated it as on-link
NTP	Removed `ntp source Loopback0`; `HQ-ASW1`, `HQ-ASW2`, `BR-ASW1` added as NTP clients	Command not available in Packet Tracer; access switches were never synchronized
Servers & endpoints	New section 6.9 with HQ-SRV1, ISP-WEB and PC GUI settings	Phase 3 depended on settings the workbook never listed
Cabling & hardware	`ISP-R1 G0/0/0` ↔ `HQ-R1 G0/0/0` is Copper Cross-Over; DCE click-order note	Router-to-router copper needs cross-over
Trunks & VLANs	Allowed VLANs pruned per access switch; access-switch VLAN databases trimmed to what each switch carries	Workbook stated pruning but trunks carried every VLAN
ACL_RESTRICT_HR	Explicit DHCP broadcast permit added; dead `eq bootpc` line removed	DHCP only worked through the final `permit ip any any`; remarks did not match behaviour

Area	Change	Reason
HSRP / OSPF alignment	Design note on the VLAN 30 return path (Section 4)	Return traffic for VLAN 30 enters at HQ-DSW1 and crosses Po1
Hardening & paste-safety	`line vty 0 15` on all routers; banners use `#` delimiter; PT notes for RSA key syntax and 3650 trunk encapsulation; production note on phone-port security	Consistent VTY policy; `^C` pasted as text left a stray “C” in every banner
Workbook Specification	Label corrected to “CCNA 200-301 Blueprint”	Previous label named the wrong certification

Appendix A is a complete test and verification checklist for the finished lab.

1. Topology Hierarchy & Hardware Inventory

The topology follows Cisco Enterprise Campus Design principles by separating functional tiers into a Dual-WAN ISP Edge, a collapsed Core/Distribution layer using redundant Catalyst 3650 multilayer switches, a dual-homed Access layer using Catalyst 2960 switches, and a remote Branch site connected via an OSPF Area 1 /30 WAN link. Before powering on modular hardware in Packet Tracer, install the required power supplies and WAN/Ethernet expansion modules listed below.

Tier	Hostname	Packet Tracer Model	Hardware / Module Notes	Primary Role & CCNA Domains Covered
WAN / ISP Sim	`ISP-R1`	Cisco 4331 (`ISR4331`)	Install `NIM-2T` in Slot 1 (`S0/1/0`)	Simulates public internet, public DMZ LAN, and dual WAN links (Gigabit primary, Serial backup); routes the HQ public NAT block `203.0.113.64/29`
HQ WAN Edge	`HQ-R1`	Cisco 4331 (`ISR4331`)	Install `NIM-2T` in Slot 1 (`S0/1/0`) & `NIM-ES2-4` in Slot 2 (`G0/2/0`–`G0/2/3`) — see note below	OSPFv2 Area 0/1 ABR & ASBR, Default Route origination, Primary/Floating Static WAN routes, Pool-based NAT/PAT, NTP Master
Branch Edge	`BR-R1`	Cisco 2911 (`ISR2911`)	Built-in `G0/0` – `G0/2`	OSPFv2 Area 1 router, Router-on-a-Stick (802.1Q subinterfaces), Local IOS DHCP Server, VTY ACLs
HQ Core / Dist	`HQ-DSW1`	Cisco 3650-24PS	Insert `AC-Power-Supply` in power slot	Layer 3 Multilayer Switch, OSPFv2 Area 0, HSRP Active (VLANs 10, 20, 50, 99) / Standby (VLAN 30), STP Root Primary (10, 20, 50, 99), DHCP Relay, Extended ACL
HQ Core / Dist	`HQ-DSW2`	Cisco 3650-24PS	Insert `AC-Power-Supply` in power slot	Layer 3 Multilayer Switch, OSPFv2 Area 0, HSRP Active (VLAN 30) / Standby (VLANs 10, 20, 50, 99), STP Root Primary (VLAN 30), DHCP Relay, Extended ACL
HQ Access 1	`HQ-ASW1`	Cisco 2960-24TT	Default fixed configuration	Layer 2 Access Switch (Engineering Data VLAN 10 & Voice VLAN 20), LACP Uplinks (`Po2`/`Po3`), Rapid PVST+, Port Security
HQ Access 2	`HQ-ASW2`	Cisco 2960-24TT	Default fixed configuration	Layer 2 Access Switch (HR/Sales VLAN 30 & Server VLAN 50), LACP Uplinks (`Po4`/`Po5`), Rapid PVST+, Port Security
Branch Access	`BR-ASW1`	Cisco 2960-24TT	Default fixed configuration	Layer 2 Branch Switch, 802.1Q Trunk to `BR-R1`, Port Security, Rapid PVST+ Root for Branch VLANs
HQ Server	`HQ-SRV1`	Server-PT	Static IP `10.10.50.10/24`	Centralized Campus DHCP Server (VLANs 10, 20, 30), Internal HTTP Web Server (Static NAT target `203.0.113.66`)
Public Server	`ISP-WEB`	Server-PT	Static IP `203.0.113.200/25`	External Public Web/DNS Server to validate PAT Overload and Extended ACL rules

Tier	Hostname	Packet Tracer Model	Hardware / Module Notes	Primary Role & CCNA Domains Covered
Endpoints	`HQ-PH1`, `HQ-PC1`, `HQ-PC2`, `BR-PC1`	IP Phone 7960 & PC-PT	Insert `IP-PHONE-POWER-SUPPLY` on `HQ-PH1`	`HQ-PC1` daisy-chained through `HQ-PH1` on `HQ-ASW1 F0/1`; `HQ-PC2` on `HQ-ASW2 F0/1`; `BR-PC1` on `BR-ASW1 F0/1`

> **Packet Tracer note — NIM-ES2-4:** The `NIM-ES2-4` is a 4-port switch module, so `G0/2/0–G0/2/3` are Layer 2 switchports and reject `ip address`. The Branch link plugs into `G0/2/0` (VLAN 1), and HQ-R1 routes it through `interface Vlan1` (`172.16.1.1/30`). Power the router off before inserting the module, then back on.

2. Physical Cabling & Port-Channel Interconnection Matrix

To prevent LACP bundle suspension between the Gigabit Ethernet ports on the Catalyst 3650 distribution switches and the FastEthernet ports on the Catalyst 2960 access switches, all physical member interfaces in Port-Channels Po2 through Po5 are explicitly locked to `speed 100` and `duplex full`. On the Serial WAN link, `ISP-R1 S0/1/0` serves as the DCE side providing a `128000` bps clock rate. In Packet Tracer the DCE end is the end you click first with the Serial DCE cable: click `ISP-R1` first, then confirm with `show controllers Serial0/1/0`. Router-to-router and switch-to-switch copper links use Cross-Over cables.

Source Device	Source Port(s)	Dest Device	Dest Port(s)	Cable Type	Logical Configuration & Function
ISP-R1	`G0/0/0`	HQ-R1	`G0/0/0`	Copper Cross-Over	Primary WAN Link (/30), Primary Static Default Route (AD 1), NAT Outside
ISP-R1	`S0/1/0` (DCE)	HQ-R1	`S0/1/0` (DTE)	Serial DCE/DTE	Backup WAN Link (/30), Floating Static Default Route (AD 5), NAT Outside
ISP-R1	`G0/0/1`	ISP-WEB	`FastEthernet0`	Copper Cross-Over	External Public DMZ (`203.0.113.128/25`) hosting ISP-WEB (`.200`)
HQ-R1	`G0/0/1`	HQ-DSW1	`G1/0/1`	Copper Straight	Routed Point-to-Point (`no switchport`), OSPFv2 Area 0 (Primary Cost 10), NAT Inside
HQ-R1	`G0/0/2`	HQ-DSW2	`G1/0/1`	Copper Straight	Routed Point-to-Point (`no switchport`), OSPFv2 Area 0 (Tuned Cost 20), NAT Inside
HQ-R1	`G0/2/0` (NIM-ES2-4)	BR-R1	`G0/0`	Copper Cross-Over	HQ-to-Branch Inter-Site WAN Transit (`172.16.1.0/30`) routed via HQ-R1 `interface Vlan1`, OSPFv2 Area 1 (broadcast), NAT Inside
HQ-DSW1	`G1/0/23`, `G1/0/24`	HQ-DSW2	`G1/0/23`, `G1/0/24`	Copper Cross-Over	`Po1` (LACP Active) — 802.1Q Trunk (Native VLAN 999), VLANs 10, 20, 30, 50, 99, 100; carries HSRP & VLAN 100 OSPF SVI
HQ-DSW1	`G1/0/2`, `G1/0/3`	HQ-ASW1	`F0/21`, `F0/22`	Copper Cross-Over	`Po2` (LACP Active) — 100 Mbps Full-Duplex 802.1Q Trunk (VLANs 10, 20, 99)
HQ-DSW2	`G1/0/2`, `G1/0/3`	HQ-ASW1	`F0/23`, `F0/24`	Copper Cross-Over	`Po3` (LACP Active) — 100 Mbps Full-Duplex 802.1Q Trunk (VLANs 10, 20, 99)
HQ-DSW1	`G1/0/4`, `G1/0/5`	HQ-ASW2	`F0/21`, `F0/22`	Copper Cross-Over	`Po4` (LACP Active) — 100 Mbps Full-Duplex 802.1Q Trunk (VLANs 30, 50, 99)

Source Device	Source Port(s)	Dest Device	Dest Port(s)	Cable Type	Logical Configuration & Function
HQ-DSW2	`G1/0/4`, `G1/0/5`	HQ-ASW2	`F0/23`, `F0/24`	Copper Cross-Over	`Po5` (LACP Active) — 100 Mbps Full-Duplex 802.1Q Trunk (VLANs 30, 50, 99)
HQ-ASW1	`F0/1`	HQ-PH1 → HQ-PC1	SW → PC Port	Copper Straight	Access VLAN 10 (Data) + Voice VLAN 20, PortFast, BPDU Guard, Port Security (Max 5 Sticky MACs)
HQ-ASW2	`F0/1`	HQ-PC2	`FastEthernet0`	Copper Straight	Access VLAN 30 (HR/Sales), PortFast, BPDU Guard, Port Security (Max 1 Sticky MAC, Shutdown)
HQ-ASW2	`F0/10`	HQ-SRV1	`FastEthernet0`	Copper Straight	Access VLAN 50 (Servers), PortFast, BPDU Guard
BR-R1	`G0/1`	BR-ASW1	`G0/1`	Copper Straight	802.1Q RoAS Trunk (`G0/1.110` Data, `G0/1.199` Mgmt, `G0/1.999` Native)
BR-ASW1	`F0/1`	BR-PC1	`FastEthernet0`	Copper Straight	Access VLAN 110 (Branch Users), PortFast, BPDU Guard, Port Security (Max 1 Sticky MAC)

3. Collision-Free IPv4 Addressing & Subnet Blueprint

Internal user, voice, server, and management networks are allocated from `10.0.0.0/8` (`10.10.x.0/24` for Headquarters and `10.20.x.0/24` for Branch), allowing a single summarized ACL line (`10.0.0.0 0.255.255.255`) to match all internal hosts for PAT overload. Point-to-point Layer 3 transit links and OSPF Router-ID loopbacks use `172.16.0.0/16`. HQ's public NAT addresses come from a dedicated block, `203.0.113.64/29`, that the ISP routes to `HQ-R1` over either WAN link, so no translation depends on a single WAN interface.

Network / Link Description	Subnet / Prefix	Subnet Mask	Device / Interface IPs	Notes & Routing Role
ISP Primary WAN Transit	`203.0.113.0/30`	`255.255.255.252`	ISP-R1 G0/0/0: `.1` HQ-R1 G0/0/0: `.2`	Primary WAN (AD 1), NAT Outside
ISP Backup Serial Transit	`198.51.100.0/30`	`255.255.255.252`	ISP-R1 S0/1/0: `.1` HQ-R1 S0/1/0: `.2`	Floating Static Backup WAN (AD 5), NAT Outside, Clock Rate 128000
ISP External Public DMZ	`203.0.113.128/25`	`255.255.255.128`	ISP-R1 G0/0/1: `.129` ISP-WEB: `.200`	Public Web/DNS Server (ISP-WEB)
HQ Public NAT Block	`203.0.113.64/29`	`255.255.255.248`	HQ_PAT pool: `.65` HQ-SRV1 Static NAT: `.66`	Routed by ISP-R1 to HQ-R1 via primary (AD 1) & floating (AD 5) statics; not bound to any interface, so NAT survives WAN failover
OSPF Transit: HQ-R1 ↔ DSW1	`172.16.0.0/30`	`255.255.255.252`	HQ-R1 G0/0/1: `.1` HQ-DSW1 G1/0/1: `.2`	OSPF Area 0 Point-to-Point (Cost 10 Primary Path)
OSPF Transit: HQ-R1 ↔ DSW2	`172.16.0.4/30`	`255.255.255.252`	HQ-R1 G0/0/2: `.5` HQ-DSW2 G1/0/1: `.6`	OSPF Area 0 Point-to-Point (Tuned Cost 20 Secondary Path)

Network / Link Description	Subnet / Prefix	Subnet Mask	Device / Interface IPs	Notes & Routing Role
OSPF Transit: DSW1 ↔ DSW2	`172.16.0.8/30`	`255.255.255.252`	HQ-DSW1 Vlan100: `.9` HQ-DSW2 Vlan100: `.10`	Dedicated L3 SVI Transit over Po1 in OSPF Area 0 (broadcast, DR/BDR)
OSPF Transit: HQ-R1 ↔ BR-R1	`172.16.1.0/30`	`255.255.255.252`	HQ-R1 Vlan1 (G0/2/0): `.1` BR-R1 G0/0: `.2`	OSPF Area 1 Inter-Site WAN Link (broadcast network type)
OSPF Router-ID Loopbacks	`172.16.255.0/24`	`255.255.255.255`	HQ-R1 Lo0: `.1/32` HQ-DSW1 Lo0: `.2/32` HQ-DSW2 Lo0: `.3/32` BR-R1 Lo0: `.4/32`	Deterministic OSPF Router-IDs; NTP server address for all clients (`172.16.255.1`)
HQ VLAN 10 (Engineering)	`10.10.10.0/24`	`255.255.255.0`	HSRP VIP: `.1` DSW1: `.2` / DSW2: `.3` DHCP: `.50–.200`	OSPF Area 0 Passive Interface, DHCP Relay to `10.10.50.10`
HQ VLAN 20 (VoIP Voice)	`10.10.20.0/24`	`255.255.255.0`	HSRP VIP: `.1` DSW1: `.2` / DSW2: `.3` DHCP: `.50–.200`	OSPF Area 0 Passive Interface, DHCP Option 150 (`10.10.50.10`)
HQ VLAN 30 (HR & Sales)	`10.10.30.0/24`	`255.255.255.0`	HSRP VIP: `.1` DSW1: `.2` / DSW2: `.3` DHCP: `.50–.200`	Filtered by inbound Extended ACL `ACL_RESTRICT_HR`
HQ VLAN 50 (Server Farm)	`10.10.50.0/24`	`255.255.255.0`	HSRP VIP: `.1` DSW1: `.2` / DSW2: `.3` HQ-SRV1: `.10`	Hosts HQ-SRV1 (DHCP & HTTP); Static NAT to `203.0.113.66`
HQ VLAN 99 (Management)	`10.10.99.0/24`	`255.255.255.0`	HSRP VIP: `.1` DSW1: `.2` / DSW2: `.3` ASW1: `.11` / ASW2: `.12`	HQ Switch Management SVIs; permitted in `ACL_VTY_MGMT`
Branch VLAN 110 (Data)	`10.20.110.0/24`	`255.255.255.0`	BR-R1 G0/1.110: `.1` DHCP: `.50–.254`	OSPF Area 1 Passive Interface; Local IOS DHCP Pool on BR-R1
Branch VLAN 199 (Mgmt)	`10.20.199.0/24`	`255.255.255.0`	BR-R1 G0/1.199: `.1` BR-ASW1 Vlan199: `.11`	Branch Switch Management SVI; permitted in `ACL_VTY_MGMT`

4. VLAN, HSRP v2, & Rapid PVST+ Root Bridge Alignment

To prevent suboptimal Layer 2 switching across the inter-distribution trunk (Po1), the Rapid PVST+ Root Bridge is deterministically synchronized with the HSRP Active router for every VLAN. **HQ-DSW1** serves as the Primary Root Bridge (priority `24576`) and HSRP Active Gateway (priority `110 preempt`) for VLANs 10, 20, 50, and 99, while **HQ-DSW2** takes primary ownership of VLAN 30.

> **Design note — VLAN 30 return path:** The alignment covers upstream (host-to-gateway) traffic. Downstream, `HQ-R1` reaches every campus subnet through `HQ-DSW1` because its uplink cost (10) is lower than `HQ-DSW2`'s (20). Return traffic for VLAN 30 therefore enters at HQ-DSW1 and crosses Po1 to HQ-DSW2, the VLAN 30 root, before reaching `HQ-ASW2`. This is an accepted trade-off at gigabit speed: the path is asymmetric but works, and the inbound ACL is applied on both switches.

VLAN	VLAN Name	Site	Subnet	Gateway VIP	HSRP Priority (DSW1 / DSW2)	STP Primary (24576)	STP Secondary (28672)
10	ENG_DATA	HQ	10.10.10.0/24	10.10.10.1	Grp 10 (110 P / 100)	HQ-DSW1	HQ-DSW2
20	VOIP_VOICE	HQ	10.10.20.0/24	10.10.20.1	Grp 20 (110 P / 100)	HQ-DSW1	HQ-DSW2
30	HR_SALES	HQ	10.10.30.0/24	10.10.30.1	Grp 30 (100 / 110 P)	HQ-DSW2	HQ-DSW1
50	SERVERS	HQ	10.10.50.0/24	10.10.50.1	Grp 50 (110 P / 100)	HQ-DSW1	HQ-DSW2
99	MGMT	HQ	10.10.99.0/24	10.10.99.1	Grp 99 (110 P / 100)	HQ-DSW1	HQ-DSW2
100	L3_TRANSIT	HQ	172.16.0.8/30	N/A (OSPF)	N/A	HQ-DSW1	HQ-DSW2
110	BR_DATA	Branch	10.20.110.0/24	10.20.110.1	N/A (RoAS)	BR-ASW1	N/A
199	BR_MGMT	Branch	10.20.199.0/24	10.20.199.1	N/A (RoAS)	BR-ASW1	N/A
888	BLACKHOLE	Both	Unassigned	None	N/A	N/A	Unused ports parked & shut
999	NATIVE_TRUNK	Both	Unassigned	None	N/A	HQ-DSW1 / BR-ASW1	HQ-DSW2 (802.1Q Native)

5. Four-Phase Implementation & Verification Guide

Phase 1: Physical Topology & Layer 2 Foundations

- **Baseline Setup & VTP Transparent Mode:** Configure hostnames, disable DNS lookup (`no ip domain-lookup`), apply MOTD banners (delimiter `#`), and set all five switches to `vtp mode transparent`.
- **VLAN Provisioning:** Create VLANs `10, 20, 30, 50, 99, 100, 888, 999` on `HQ-DSW1` and `HQ-DSW2`; create only the VLANs each access switch carries — `10, 20, 99, 888, 999` on `HQ-ASW1` and `30, 50, 99, 888, 999` on `HQ-ASW2`; create `110, 199, 888, 999` on `BR-ASW1`.
- **LACP EtherChannels & 802.1Q Trunks:** Bundle physical links into `Po1`–`Po5` using active LACP (`channel-group <1-5> mode active`), enforce static 802.1Q trunking with `switchport nonegotiate`, set Native VLAN to `999`, and prune allowed VLANs: `Po1` → `10,20,30,50,99,100,999`; `Po2`/`Po3` → `10,20,99,999`; `Po4`/`Po5` → `30,50,99,999`.
- **Edge Access Ports & Blackhole Parking:** Configure Data/Voice access ports with `spanning-tree portfast` and `spanning-tree bpduguard enable`; assign all unused switchports to `vlan 888` and issue `shutdown`.
- **Rapid PVST+ & HSRP v2 Alignment:** Enable `spanning-tree mode rapid-pvst` and configure HSRP v2 groups `10, 20, 30, 50, 99` on `HQ-DSW1` and `HQ-DSW2` so STP Root Bridges match HSRP Active gateways.

> **Packet Tracer note — 3650 trunks:** If `switchport mode trunk` on a 3650 port-channel returns "An interface whose trunk encapsulation is Auto can not be configured to trunk mode", enter `switchport trunk encapsulation dot1q` first, then repeat the command.

Phase 1 Verification Command	Target Device(s)	Expected Verification Output Signals
`show vtp status`	All Switches	`VTP Operating Mode : Transparent` across all campus and branch switches.
`show vlan brief`	All Switches	Only the provisioned VLANs listed per switch; access ports mapped to 10/20, 30, 50, 110; all unused ports parked in `888 (BLACKHOLE)`.
`show etherchannel summary`	HQ-DSW1, HQ-DSW2, HQ-ASW1, HQ-ASW2	Flag `(SU)` next to `Po1–Po5` (Layer 2, In-use) and `(P)` next to every physical member port (Protocol: LACP).
`show interfaces trunk`	All Switches	`Po1–Po5` and `BR-ASW1 G0/1` showing `802.1q` encapsulation, status: `trunking`, `Native vlan: 999`, and allowed VLANs matching the pruned lists above.
`show spanning-tree summary`	All Switches	Switch operating in `rapid-pvst` mode; `HQ-DSW1` is Root for VLANs 10, 20, 50, 99 and `HQ-DSW2` is Root for VLAN 30.
`show standby brief`	HQ-DSW1, HQ-DSW2	Column `P` confirms preemption; `HQ-DSW1` is Active (Priority 110) for Groups 10, 20, 50, 99 and Standby for Group 30; `HQ-DSW2` is Active for Group 30.

Phase 2: Layer 3 Core, Routing & WAN

- **Routed Ports, Subinterfaces, & Loopbacks:** Configure `no switchport` on `HQ-DSW1/DSW2 G1/0/1`, configure `interface Vlan100` over `Po1`, route the Branch link through `interface Vlan1` on HQ-R1 (its NIM-ES2-4 ports are switchports), build 802.1Q RoAS subinterfaces on `BR-R1 G0/1`, and assign `Loopback0` (`172.16.255.1–.4/32`).
- **Multi-Area OSPFv2:** Configure `router ospf 1` with deterministic `router-id` values, advertise HQ core/campus networks in Area 0 and Branch transit/LANs in Area 1, apply `passive-interface default`, set `ip ospf network point-to-point` on the routed HQ-R1 ↔ DSW uplinks (the VLAN-interface links — HQ-R1 `Vlan1` ↔ BR-R1 and `Vlan100` between the DSWs — stay broadcast, because Packet Tracer ignores the command on VLAN interfaces), and tune `HQ-R1` ↔ `HQ-DSW2` to `ip ospf cost 20`.
- **WAN Redundancy & Default Origination:** Configure primary (`AD 1` via `203.0.113.1`) and floating static (`AD 5` via `198.51.100.1`) default routes on `HQ-R1` and inject into OSPF via `default-information originate`. On `ISP-R1`, route the HQ public NAT block `203.0.113.64/29` back to HQ over the same primary/floating pair.

Phase 2 Verification Command	Target Device(s)	Expected Verification Output Signals
`show ip interface brief`	All Routers & L3 Switches	Every interface in use (physical, subinterface, loopback, SVI and HQ-R1 `Vlan1`) shows `up` / `up`; unused ports show `administratively down`.
`show ip ospf neighbor`	HQ-R1, HQ-DSW1, HQ-DSW2, BR-R1	`FULL/ -` on the point-to-point uplinks (HQ-R1 ↔ DSW1/DSW2); `FULL/DR` or `FULL/BDR` on the broadcast links (HQ-R1 ↔ BR-R1 over `Vlan1`, DSW1 ↔ DSW2 over `Vlan100`). HQ-R1 lists three neighbors: `172.16.255.2`, `.3`, `.4`.
`show ip protocols`	HQ-R1, HQ-DSW1, BR-R1	Confirms Routing Protocol is "ospf 1", Router IDs, and lists all end-user SVIs/subinterfaces under Passive Interface(s).

Phase 2 Verification Command	Target Device(s)	Expected Verification Output Signals
`show ip route ospf`	HQ-DSW1, BR-R1	Displays intra-area (`O`), inter-area (`O IA` between Areas 0 and 1), and external default route (`O*E2 0.0.0.0/0 [110/1]`).
`show ip route static`	HQ-R1, ISP-R1	HQ-R1 shows `S* 0.0.0.0/0 [1/0] via 203.0.113.1`; ISP-R1 shows `S 203.0.113.64/29 [1/0] via 203.0.113.2`. Shutting down `HQ-R1 G0/0/0` fails both over to the `[5/0]` routes via `198.51.100.x`.
`ping 203.0.113.200` (with `HQ-R1 G0/0/0` shut)	HQ-PC1, BR-PC1	Succeeds over the Serial backup — PAT and Static NAT keep working because the public block is reachable through either WAN link. Run after Phase 3.

Phase 3: Infrastructure Services (DHCP, NAT/PAT, & NTP)

- **Centralized & Local DHCP:** Configure DHCP pools on `HQ-SRV1` (`10.10.50.10`) for VLANs 10, 20 (with Option 150 TFTP `10.10.50.10`), and 30 — full GUI settings in Section 6.9; add `ip helper-address 10.10.50.10` on both `HQ-DSW1` and `HQ-DSW2` SVIs; configure local IOS DHCP pool `BR_DATA_POOL` on `BR-R1`.
- **Static NAT & PAT Overload:** Mark inside/outside NAT interfaces on `HQ-R1`, map `10.10.50.10` statically to `203.0.113.66`, and overload `NAT_ELIGIBLE` (`10.0.0.0/8`) onto NAT pool `HQ_PAT` (`203.0.113.65`). Both addresses come from `203.0.113.64/29`, which ISP-R1 routes to HQ over either WAN link, so translations survive a failover to the Serial backup.
- **NTP Synchronization:** Set `HQ-R1` as `ntp master 2` and point `HQ-DSW1`, `HQ-DSW2`, `BR-R1`, `HQ-ASW1`, `HQ-ASW2`, and `BR-ASW1` at its Loopback0 with `ntp server 172.16.255.1`. The access switches reach it through their management SVI gateways.

> **Packet Tracer note — DHCP stuck after many changes:** If PCs get no lease even though every configuration checks out, save the lab, close Packet Tracer completely and reopen it, then renew the leases. Packet Tracer can keep a stale DHCP relay state that only a restart clears.

> **Packet Tracer note — NTP:** `ntp source` is not available in Packet Tracer, and it is not needed here: HQ-R1 replies from the Loopback0 address the clients query.

Phase 3 Verification Command	Target Device(s)	Expected Verification Output Signals
`ipconfig /all`	HQ-PC1, HQ-PC2, BR-PC1	Endpoints receive `.50+` leases in their respective subnets with `.1` default gateways and DNS `203.0.113.200`.
`show ip dhcp binding`	BR-R1	Displays an Automatic lease binding (`10.20.110.50` or higher) to BR-PC1's MAC address.
`show ip nat translations`	HQ-R1	Displays permanent static mapping `203.0.113.66` ↔ `10.10.50.10` and dynamic PAT entries with inside global `203.0.113.65` when internal hosts ping `203.0.113.200`.
`show ntp status`	HQ-DSW1, HQ-DSW2, BR-R1, HQ-ASW1, HQ-ASW2, BR-ASW1	`Clock is synchronized, stratum 3, reference is 172.16.255.1`.

Phase 4: Network Security & Mitigation

- **Layer 2 Edge Port Security:** Enable sticky MAC learning with `maximum 5` and `violation restrict` on `HQ-ASW1 F0/1` (IP Phone + PC, with headroom because the phone's MAC is learned in both VLANs), and `maximum 1` with `violation shutdown` on `HQ-ASW2 F0/1` and `BR-ASW1 F0/1`.
- **Inter-VLAN & VTY Access Control Lists:** Apply Extended Named ACL `ACL_RESTRICT_HR` inbound on `interface Vlan30` (`HQ-DSW1` & `HQ-DSW2`) to permit only DHCP (client broadcasts and unicast renewals) and HTTP to HQ-SRV1 while blocking other traffic to Server/Management VLANs; apply `ACL_VTY_MGMT` to all VTY lines.

> **Packet Tracer note — DHCP Snooping & DAI:** These are not part of this lab. With DHCP Snooping enabled, Packet Tracer 9.0.1 dropped the relayed DHCP replies to the HQ PCs, so they got no lease. DAI needs the snooping binding table, so it was removed too. Both remain CCNA exam topics; practise them in a smaller lab without DHCP relay, or on real switches.

Phase 4 Verification Command	Target Device(s)	Expected Verification Output Signals
`show port-security interface F0/1`	HQ-ASW1, HQ-ASW2, BR-ASW1	`Port Status : Secure-up`, `Maximum MAC Addresses : 5` on HQ-ASW1 and `1` on HQ-ASW2/BR-ASW1, `Violation Count : 0`.
`show access-lists`	HQ-DSW1, HQ-DSW2, All Devices	Hit counters increment on `ACL_RESTRICT_HR` (permitting HTTP/DHCP, denying ICMP/SSH from VLAN 30 to VLAN 50/99) and `ACL_VTY_MGMT`.

6. Complete Device-by-Device Cisco IOS Startup Configurations

Each configuration block below is complete and ready to paste directly into the privileged EXEC prompt (`enable`) of its corresponding router or switch in Cisco Packet Tracer. All commands include full interface expansions, inline explanatory comments (`!`), and baseline hardening.

6.1. WAN / ISP SIMULATOR ROUTER: ISP-R1 (Cisco ISR 4331)

Architectural Role & Key Functions: Simulates Public Internet, Dual WAN Termination, Public Routing, and return routing for the HQ public NAT block

---

```
enable
configure terminal
! --- Baseline Device Hardening ---
hostname ISP-R1
no ip domain-lookup
ip domain-name isp.lab
enable secret CiscoCCNA#Enable1
service password-encryption
username admin privilege 15 secret CiscoCCNA#Admin1
! Banner uses '#' as the delimiter so pasted text does not leave a stray "C"
banner motd #
**************************************************************************
* UNAUTHORIZED ACCESS TO THIS NETWORK DEVICE IS PROHIBITED.              *
* All connections are monitored and logged. Disconnect immediately       *
* if you are not an authorized administrator.                            *
**************************************************************************
#
! Generate 2048-bit RSA keys and enforce SSHv2
! (If PT rejects the one-line form, enter "crypto key generate rsa" and answer 2048)
crypto key generate rsa general-keys modulus 2048
ip ssh version 2
ip ssh time-out 60
ip ssh authentication-retries 3
! --- WAN & Public DMZ Interfaces ---
! Primary Gigabit WAN Link to HQ-R1 (G0/0/0) - Copper Cross-Over cable
interface GigabitEthernet0/0/0
 description PRIMARY_WAN_TO_HQ-R1_G0/0/0
 ip address 203.0.113.1 255.255.255.252
 no shutdown
 exit
! External Public Server LAN (Hosts ISP-WEB / DNS at 203.0.113.200)
interface GigabitEthernet0/0/1
 description PUBLIC_DMZ_TO_ISP-WEB
 ip address 203.0.113.129 255.255.255.128
 no shutdown
 exit
! Unused Built-in Gigabit Interface
interface GigabitEthernet0/0/2
 description UNUSED_PORT
 shutdown
 exit
! Backup Serial WAN Link to HQ-R1 (S0/1/0 - DCE Side provides clock rate)
! In PT the DCE end is the end clicked first with the Serial DCE cable
interface Serial0/1/0
 description BACKUP_WAN_TO_HQ-R1_S0/1/0
 ip address 198.51.100.1 255.255.255.252
 clock rate 128000
 no shutdown
 exit
! Unused Serial Interface
interface Serial0/1/1
 description UNUSED_PORT
 shutdown
 exit
! --- Static Return Routing for the HQ Public NAT Block (203.0.113.64/29) ---
```

```
! Covers the HQ PAT pool address (.65) and HQ-SRV1 Static NAT address (.66)
! Primary return route via the Gigabit WAN link
ip route 203.0.113.64 255.255.255.248 203.0.113.2
! Floating static return route (AD 5) via the Serial backup WAN link
ip route 203.0.113.64 255.255.255.248 198.51.100.2 5
! --- Console & VTY Line Hardening ---
line console 0
 logging synchronous
 exec-timeout 5 0
 login local
 exit
line vty 0 15
 logging synchronous
 exec-timeout 5 0
 transport input ssh
 login local
 exit
end
write memory
```

6.2. HQ WAN / EDGE ROUTER: HQ-R1 (Cisco ISR 4331)

Architectural Role & Key Functions: OSPF ABR (Area 0 / Area 1) & ASBR, Pool-based NAT/PAT Edge, Primary/Floating Static Default Routes, and Authoritative Stratum 2 NTP Master

---

```
enable
configure terminal
! --- Baseline Device Hardening ---
hostname HQ-R1
no ip domain-lookup
ip domain-name ccna.lab
enable secret CiscoCCNA#Enable1
service password-encryption
username admin privilege 15 secret CiscoCCNA#Admin1
! Banner uses '#' as the delimiter so pasted text does not leave a stray "C"
banner motd #
**************************************************************************
* UNAUTHORIZED ACCESS TO THIS NETWORK DEVICE IS PROHIBITED.              *
* HQ-R1 Edge Router - Authorized Personnel Only.                         *
**************************************************************************
#
! Generate 2048-bit RSA keys and enforce SSHv2
! (If PT rejects the one-line form, enter "crypto key generate rsa" and answer 2048)
crypto key generate rsa general-keys modulus 2048
ip ssh version 2
ip ssh time-out 60
ip ssh authentication-retries 3
! --- Loopback Interface (OSPF Router-ID & NTP Server Address) ---
interface Loopback0
 description OSPF_RID_AND_NTP_SERVER_ADDRESS
 ip address 172.16.255.1 255.255.255.255
 exit
! --- Outside WAN Interfaces ---
! Primary ISP Uplink (NAT Outside) - Copper Cross-Over cable to ISP-R1
interface GigabitEthernet0/0/0
 description PRIMARY_WAN_TO_ISP-R1_G0/0/0
 ip address 203.0.113.2 255.255.255.252
 ip nat outside
 no shutdown
 exit
! Backup Serial ISP Uplink (NAT Outside)
interface Serial0/1/0
 description BACKUP_WAN_TO_ISP-R1_S0/1/0
 ip address 198.51.100.2 255.255.255.252
 ip nat outside
 no shutdown
 exit
! Unused Serial Interface
interface Serial0/1/1
 description UNUSED_PORT
 shutdown
 exit
! --- Inside Campus Core & Branch Transit Interfaces ---
! Routed Transit Link to HQ-DSW1 (OSPF Area 0 - Primary Path Cost 10)
interface GigabitEthernet0/0/1
 description L3_TRANSIT_TO_HQ-DSW1_G1/0/1
 ip address 172.16.0.1 255.255.255.252
 ip nat inside
 ip ospf network point-to-point
 ip ospf cost 10
 no shutdown
 exit
! Routed Transit Link to HQ-DSW2 (OSPF Area 0 - Secondary Path Tuned Cost 20)
interface GigabitEthernet0/0/2
 description L3_TRANSIT_TO_HQ-DSW2_G1/0/1
 ip address 172.16.0.5 255.255.255.252
 ip nat inside
 ip ospf network point-to-point
```

```
 ip ospf cost 20
 no shutdown
 exit
! --- Inter-Site WAN Link to BR-R1 via NIM-ES2-4 (Slot 2) ---
! NIM-ES2-4 ports G0/2/0 - 3 are Layer 2 switchports and cannot hold an IP address.
! G0/2/0 stays in VLAN 1 and the routed address lives on interface Vlan1.
interface GigabitEthernet0/2/0
 description INTERSITE_WAN_TO_BR-R1_G0/0
 switchport mode access
 no shutdown
 exit
! Unused NIM-ES2-4 Switchports
interface range GigabitEthernet0/2/1 - 3
 description UNUSED_PORT
 switchport mode access
 shutdown
 exit
! Routed SVI for the HQ-to-Branch Link (OSPF Area 1, NAT Inside)
! OSPF runs as broadcast here: Packet Tracer ignores "ip ospf network point-to-point" on VLAN interfaces
interface Vlan1
 description INTERSITE_WAN_TO_BR-R1_G0/0
 ip address 172.16.1.1 255.255.255.252
 ip nat inside
 no shutdown
 exit
! --- Primary & Floating Static Default Routing ---
! Primary Default Route via Gigabit ISP Link (Default AD = 1)
ip route 0.0.0.0 0.0.0.0 203.0.113.1
! Floating Static Backup Default Route via Serial ISP Link (AD = 5)
ip route 0.0.0.0 0.0.0.0 198.51.100.1 5
! --- Multi-Area OSPFv2 Configuration (ABR & ASBR) ---
router ospf 1
 router-id 172.16.255.1
 ! Suppress OSPF Hellos on all interfaces by default for security
 passive-interface default
 ! Enable OSPF adjacencies strictly on internal transit links
 no passive-interface GigabitEthernet0/0/1
 no passive-interface GigabitEthernet0/0/2
 no passive-interface Vlan1
 ! Area 0 (Backbone) Networks
 network 172.16.255.1 0.0.0.0 area 0
 network 172.16.0.0 0.0.0.3 area 0
 network 172.16.0.4 0.0.0.3 area 0
 ! Area 1 (Branch Site) Networks
 network 172.16.1.0 0.0.0.3 area 1
 ! Propagate the active default route into OSPF domain
 default-information originate
 exit
! --- NAT / PAT Configuration ---
! The HQ public block 203.0.113.64/29 is routed to HQ-R1 by ISP-R1 over BOTH
! WAN links, so translations are not tied to G0/0/0 and survive a failover
! to the Serial backup.
! Static Inside NAT publishing HQ-SRV1 (10.10.50.10) to 203.0.113.66
ip nat inside source static 10.10.50.10 203.0.113.66
! Standard Named ACL identifying internal RFC 1918 traffic eligible for PAT
ip access-list standard NAT_ELIGIBLE
 remark Permit all internal HQ and Branch 10.0.0.0/8 subnets for PAT
 permit 10.0.0.0 0.255.255.255
 exit
! Single-address PAT pool (203.0.113.65) taken from the HQ public block
ip nat pool HQ_PAT 203.0.113.65 203.0.113.65 netmask 255.255.255.248
! Bind NAT_ELIGIBLE to the pool with PAT Overload
ip nat inside source list NAT_ELIGIBLE pool HQ_PAT overload
! --- NTP Master Configuration ---
! Authoritative Stratum 2 clock. Clients point at Loopback0 (172.16.255.1).
! "ntp source" is not available in Packet Tracer and is not required here.
ntp master 2
! --- VTY Management Access Control List ---
ip access-list standard ACL_VTY_MGMT
```

```
 remark Permit SSH only from HQ and Branch Management VLANs
 permit 10.10.99.0 0.0.0.255
 permit 10.20.199.0 0.0.0.255
 deny any
 exit
! --- Console & VTY Line Hardening ---
line console 0
 logging synchronous
 exec-timeout 5 0
 login local
 exit
line vty 0 15
 access-class ACL_VTY_MGMT in
 logging synchronous
 exec-timeout 5 0
 transport input ssh
 login local
 exit
end
write memory
```

6.3. BRANCH EDGE ROUTER: BR-R1 (Cisco ISR 2911)

Architectural Role & Key Functions: OSPF Area 1 Router, Router-on-a-Stick (802.1Q Subinterfaces), Local IOS DHCP Server, and NTP Client

---

```
enable
configure terminal
! --- Baseline Device Hardening ---
hostname BR-R1
no ip domain-lookup
ip domain-name ccna.lab
enable secret CiscoCCNA#Enable1
service password-encryption
username admin privilege 15 secret CiscoCCNA#Admin1
! Banner uses '#' as the delimiter so pasted text does not leave a stray "C"
banner motd #
**************************************************************************
* UNAUTHORIZED ACCESS TO THIS NETWORK DEVICE IS PROHIBITED.              *
* BR-R1 Branch Router - Authorized Personnel Only.                       *
**************************************************************************
#
! Generate 2048-bit RSA keys and enforce SSHv2
! (If PT rejects the one-line form, enter "crypto key generate rsa" and answer 2048)
crypto key generate rsa general-keys modulus 2048
ip ssh version 2
ip ssh time-out 60
ip ssh authentication-retries 3
! --- Loopback Interface (OSPF Router-ID) ---
interface Loopback0
 description OSPF_RID
 ip address 172.16.255.4 255.255.255.255
 exit
! --- WAN Uplink to HQ-R1 ---
interface GigabitEthernet0/0
 description INTERSITE_WAN_TO_HQ-R1_G0/2/0_VLAN1
 ip address 172.16.1.2 255.255.255.252
 ! Default broadcast OSPF network type, matching HQ-R1 Vlan1 (a type mismatch stops routes installing)
 no shutdown
 exit
! --- Router-on-a-Stick (RoAS) Trunk to BR-ASW1 ---
interface GigabitEthernet0/1
 description ROAS_8021Q_TRUNK_TO_BR-ASW1_G0/1
 no ip address
 no shutdown
 exit
! Subinterface for Branch Data VLAN 110
interface GigabitEthernet0/1.110
 description BRANCH_DATA_VLAN_110_GATEWAY
 encapsulation dot1Q 110
 ip address 10.20.110.1 255.255.255.0
 exit
! Subinterface for Branch Management VLAN 199
interface GigabitEthernet0/1.199
 description BRANCH_MGMT_VLAN_199_GATEWAY
 encapsulation dot1Q 199
 ip address 10.20.199.1 255.255.255.0
 exit
! Subinterface for Unassigned Native VLAN 999
interface GigabitEthernet0/1.999
 description NATIVE_VLAN_999_UNASSIGNED
 encapsulation dot1Q 999 native
 no ip address
 exit
! Unused Built-in Interface
interface GigabitEthernet0/2
 description UNUSED_PORT
 shutdown
 exit
```

```
! --- Local Cisco IOS DHCP Server for Branch VLAN 110 ---
! Reserve .1 through .49 for static infrastructure assignments
ip dhcp excluded-address 10.20.110.1 10.20.110.49
ip dhcp pool BR_DATA_POOL
 network 10.20.110.0 255.255.255.0
 default-router 10.20.110.1
 dns-server 203.0.113.200
 domain-name ccna.lab
 exit
! --- OSPFv2 Area 1 Configuration ---
router ospf 1
 router-id 172.16.255.4
 ! Prevent OSPF Hellos from being sent toward Branch end-hosts
 passive-interface default
 no passive-interface GigabitEthernet0/0
 ! Advertise Loopback, WAN Transit, and Branch LANs into Area 1
 network 172.16.255.4 0.0.0.0 area 1
 network 172.16.1.0 0.0.0.3 area 1
 network 10.20.110.0 0.0.0.255 area 1
 network 10.20.199.0 0.0.0.255 area 1
 exit
! --- NTP Client Synchronization ---
ntp server 172.16.255.1
! --- VTY Management Access Control List ---
ip access-list standard ACL_VTY_MGMT
 remark Permit SSH only from HQ and Branch Management VLANs
 permit 10.10.99.0 0.0.0.255
 permit 10.20.199.0 0.0.0.255
 deny any
 exit
! --- Console & VTY Line Hardening ---
line console 0
 logging synchronous
 exec-timeout 5 0
 login local
 exit
line vty 0 15
 access-class ACL_VTY_MGMT in
 logging synchronous
 exec-timeout 5 0
 transport input ssh
 login local
 exit
end
write memory
```

6.4. HQ CORE / DISTRIBUTION SWITCH 1: HQ-DSW1 (Cisco Catalyst 3650-24PS)

Architectural Role & Key Functions: STP Root Primary (VLANs 10, 20, 50, 99, 100, 999), HSRP Active Gateway (VLANs 10, 20, 50, 99) & Standby (VLAN 30), OSPF Area 0, DHCP Relay, and Inter-VLAN Security ACL Enforcement

---

```
enable
configure terminal
! --- Baseline Device Hardening ---
hostname HQ-DSW1
no ip domain-lookup
ip domain-name ccna.lab
enable secret CiscoCCNA#Enable1
service password-encryption
username admin privilege 15 secret CiscoCCNA#Admin1
! Banner uses '#' as the delimiter so pasted text does not leave a stray "C"
banner motd #
**************************************************************************
* UNAUTHORIZED ACCESS TO THIS NETWORK DEVICE IS PROHIBITED.              *
* HQ-DSW1 Core/Distribution Switch 1 - Authorized Personnel Only.        *
**************************************************************************
#
! Generate 2048-bit RSA keys and enforce SSHv2
! (If PT rejects the one-line form, enter "crypto key generate rsa" and answer 2048)
crypto key generate rsa general-keys modulus 2048
ip ssh version 2
ip ssh time-out 60
ip ssh authentication-retries 3
! Enable Layer 3 Multilayer Routing and VTP Transparent Mode
ip routing
vtp mode transparent
! --- VLAN Database Provisioning ---
vlan 10
 name ENG_DATA
 exit
vlan 20
 name VOIP_VOICE
 exit
vlan 30
 name HR_SALES
 exit
vlan 50
 name SERVERS
 exit
vlan 99
 name MGMT
 exit
vlan 100
 name L3_TRANSIT
 exit
vlan 888
 name BLACKHOLE
 exit
vlan 999
 name NATIVE_TRUNK
 exit
! --- Rapid PVST+ Root Bridge Alignment ---
spanning-tree mode rapid-pvst
! Primary Root (Priority 24576) for VLANs where HQ-DSW1 is HSRP Active
spanning-tree vlan 10,20,50,99,100,999 priority 24576
! Secondary Root (Priority 28672) for VLAN 30 (where HQ-DSW2 is HSRP Active)
spanning-tree vlan 30 priority 28672
! --- Loopback & Layer 3 Routed Uplink to HQ-R1 ---
interface Loopback0
 description OSPF_RID
 ip address 172.16.255.2 255.255.255.255
 exit
```

```
interface GigabitEthernet1/0/1
 description L3_ROUTED_UPLINK_TO_HQ-R1_G0/0/1
 no switchport
 ip address 172.16.0.2 255.255.255.252
 ip ospf network point-to-point
 ip ospf cost 10
 no shutdown
 exit
! --- LACP EtherChannel Member Links & Port-Channel Trunks ---
! Po1: Inter-Switch Trunk to HQ-DSW2 (G1/0/23 - 24)
interface GigabitEthernet1/0/23
 description PO1_MEMBER_TO_HQ-DSW2_G1/0/23
 channel-group 1 mode active
 no shutdown
 exit
interface GigabitEthernet1/0/24
 description PO1_MEMBER_TO_HQ-DSW2_G1/0/24
 channel-group 1 mode active
 no shutdown
 exit
! If PT rejects trunk mode ("encapsulation is Auto"), first enter:
!   switchport trunk encapsulation dot1q
interface Port-channel1
 description LACP_TRUNK_TO_HQ-DSW2
 switchport mode trunk
 switchport trunk native vlan 999
 switchport trunk allowed vlan 10,20,30,50,99,100,999
 switchport nonegotiate
 exit
! Po2: Downlink Trunk to HQ-ASW1 (G1/0/2 - 3; speed/duplex matched to FastEthernet)
! Pruned to the VLANs HQ-ASW1 actually carries (10, 20, 99 + native 999)
interface GigabitEthernet1/0/2
 description PO2_MEMBER_TO_HQ-ASW1_F0/21
 speed 100
 duplex full
 channel-group 2 mode active
 no shutdown
 exit
interface GigabitEthernet1/0/3
 description PO2_MEMBER_TO_HQ-ASW1_F0/22
 speed 100
 duplex full
 channel-group 2 mode active
 no shutdown
 exit
interface Port-channel2
 description LACP_TRUNK_TO_HQ-ASW1
 switchport mode trunk
 switchport trunk native vlan 999
 switchport trunk allowed vlan 10,20,99,999
 switchport nonegotiate
 exit
! Po4: Downlink Trunk to HQ-ASW2 (G1/0/4 - 5; speed/duplex matched to FastEthernet)
! Pruned to the VLANs HQ-ASW2 actually carries (30, 50, 99 + native 999)
interface GigabitEthernet1/0/4
 description PO4_MEMBER_TO_HQ-ASW2_F0/21
 speed 100
 duplex full
 channel-group 4 mode active
 no shutdown
 exit
interface GigabitEthernet1/0/5
 description PO4_MEMBER_TO_HQ-ASW2_F0/22
 speed 100
 duplex full
 channel-group 4 mode active
 no shutdown
 exit
interface Port-channel4
```

```
 description LACP_TRUNK_TO_HQ-ASW2
 switchport mode trunk
 switchport trunk native vlan 999
 switchport trunk allowed vlan 30,50,99,999
 switchport nonegotiate
 exit
! --- Park All Unused Physical Ports in BLACKHOLE VLAN 888 & Disable ---
interface range GigabitEthernet1/0/6 - 22, GigabitEthernet1/1/1 - 4
 description UNUSED_BLACKHOLE_PORT
 switchport mode access
 switchport access vlan 888
 shutdown
 exit
! --- Inter-VLAN Security Extended Named ACL ---
! Restricts HR/Sales (VLAN 30) from the Server Farm (except DHCP/HTTP to HQ-SRV1)
! and blocks all access from VLAN 30 to Campus/Branch Management VLANs
ip access-list extended ACL_RESTRICT_HR
 remark Permit DHCP Discover/Request broadcasts (relayed by ip helper-address)
 permit udp host 0.0.0.0 host 255.255.255.255 eq bootps
 remark Permit DHCP unicast renewals to HQ-SRV1 (client port 68 to server port 67)
 permit udp 10.10.30.0 0.0.0.255 host 10.10.50.10 eq bootps
 remark Permit HTTP Web Access to HQ-SRV1
 permit tcp 10.10.30.0 0.0.0.255 host 10.10.50.10 eq www
 remark Block all other access to Server Farm VLAN 50
 deny ip 10.10.30.0 0.0.0.255 10.10.50.0 0.0.0.255
 remark Block all access to HQ Management VLAN 99 and Branch Management VLAN 199
 deny ip 10.10.30.0 0.0.0.255 10.10.99.0 0.0.0.255
 deny ip 10.10.30.0 0.0.0.255 10.20.199.0 0.0.0.255
 remark Permit all other corporate and internet traffic
 permit ip any any
 exit
! --- Switched Virtual Interfaces (SVIs), HSRP v2, & DHCP Relay ---
! VLAN 10: Engineering Data (HSRP Active - Priority 110 with Preemption)
interface Vlan10
 description ENG_DATA_SVI_HSRP_ACTIVE
 ip address 10.10.10.2 255.255.255.0
 ip helper-address 10.10.50.10
 standby version 2
 standby 10 ip 10.10.10.1
 standby 10 priority 110
 standby 10 preempt
 no shutdown
 exit
! VLAN 20: VoIP Voice (HSRP Active - Priority 110 with Preemption)
interface Vlan20
 description VOIP_VOICE_SVI_HSRP_ACTIVE
 ip address 10.10.20.2 255.255.255.0
 ip helper-address 10.10.50.10
 standby version 2
 standby 20 ip 10.10.20.1
 standby 20 priority 110
 standby 20 preempt
 no shutdown
 exit
! VLAN 30: HR & Sales (HSRP Standby - Default Priority 100; ACL Applied Inbound)
interface Vlan30
 description HR_SALES_SVI_HSRP_STANDBY
 ip address 10.10.30.2 255.255.255.0
 ip helper-address 10.10.50.10
 ip access-group ACL_RESTRICT_HR in
 standby version 2
 standby 30 ip 10.10.30.1
 standby 30 priority 100
 no shutdown
 exit
! VLAN 50: Server Farm (HSRP Active - Priority 110 with Preemption)
interface Vlan50
 description SERVERS_SVI_HSRP_ACTIVE
 ip address 10.10.50.2 255.255.255.0
```

```
 standby version 2
 standby 50 ip 10.10.50.1
 standby 50 priority 110
 standby 50 preempt
 no shutdown
 exit
! VLAN 99: Campus Management (HSRP Active - Priority 110 with Preemption)
interface Vlan99
 description MGMT_SVI_HSRP_ACTIVE
 ip address 10.10.99.2 255.255.255.0
 standby version 2
 standby 99 ip 10.10.99.1
 standby 99 priority 110
 standby 99 preempt
 no shutdown
 exit
! VLAN 100: Dedicated Layer 3 OSPF Transit SVI to HQ-DSW2 over Po1
! OSPF runs as broadcast (DR/BDR election): Packet Tracer does not accept point-to-point on SVIs
interface Vlan100
 description L3_OSPF_TRANSIT_SVI_TO_HQ-DSW2
 ip address 172.16.0.9 255.255.255.252
 no shutdown
 exit
! --- OSPFv2 Area 0 Configuration ---
router ospf 1
 router-id 172.16.255.2
 ! Suppress OSPF Hellos on all end-user/server VLAN SVIs
 passive-interface default
 no passive-interface GigabitEthernet1/0/1
 no passive-interface Vlan100
 ! Advertise Loopback, Core Transit Links, and Campus Subnets into Area 0
 network 172.16.255.2 0.0.0.0 area 0
 network 172.16.0.0 0.0.0.3 area 0
 network 172.16.0.8 0.0.0.3 area 0
 network 10.10.10.0 0.0.0.255 area 0
 network 10.10.20.0 0.0.0.255 area 0
 network 10.10.30.0 0.0.0.255 area 0
 network 10.10.50.0 0.0.0.255 area 0
 network 10.10.99.0 0.0.0.255 area 0
 exit
! --- NTP Client Synchronization ---
ntp server 172.16.255.1
! --- VTY Management Access Control List ---
ip access-list standard ACL_VTY_MGMT
 remark Permit SSH only from HQ and Branch Management VLANs
 permit 10.10.99.0 0.0.0.255
 permit 10.20.199.0 0.0.0.255
 deny any
 exit
! --- Console & VTY Line Hardening ---
line console 0
 logging synchronous
 exec-timeout 5 0
 login local
 exit
line vty 0 15
 access-class ACL_VTY_MGMT in
 logging synchronous
 exec-timeout 5 0
 transport input ssh
 login local
 exit
end
write memory
```

6.5. HQ CORE / DISTRIBUTION SWITCH 2: HQ-DSW2 (Cisco Catalyst 3650-24PS)

Architectural Role & Key Functions: STP Root Primary (VLAN 30) & Secondary (VLANs 10, 20, 50, 99, 100, 999), HSRP Active Gateway (VLAN 30) & Standby (VLANs 10, 20, 50, 99), OSPF Area 0, DHCP Relay, and Inter-VLAN Security ACL Enforcement

---

```
enable
configure terminal
! --- Baseline Device Hardening ---
hostname HQ-DSW2
no ip domain-lookup
ip domain-name ccna.lab
enable secret CiscoCCNA#Enable1
service password-encryption
username admin privilege 15 secret CiscoCCNA#Admin1
! Banner uses '#' as the delimiter so pasted text does not leave a stray "C"
banner motd #
**************************************************************************
* UNAUTHORIZED ACCESS TO THIS NETWORK DEVICE IS PROHIBITED.              *
* HQ-DSW2 Core/Distribution Switch 2 - Authorized Personnel Only.        *
**************************************************************************
#
! Generate 2048-bit RSA keys and enforce SSHv2
! (If PT rejects the one-line form, enter "crypto key generate rsa" and answer 2048)
crypto key generate rsa general-keys modulus 2048
ip ssh version 2
ip ssh time-out 60
ip ssh authentication-retries 3
! Enable Layer 3 Multilayer Routing and VTP Transparent Mode
ip routing
vtp mode transparent
! --- VLAN Database Provisioning ---
vlan 10
 name ENG_DATA
 exit
vlan 20
 name VOIP_VOICE
 exit
vlan 30
 name HR_SALES
 exit
vlan 50
 name SERVERS
 exit
vlan 99
 name MGMT
 exit
vlan 100
 name L3_TRANSIT
 exit
vlan 888
 name BLACKHOLE
 exit
vlan 999
 name NATIVE_TRUNK
 exit
! --- Rapid PVST+ Root Bridge Alignment ---
spanning-tree mode rapid-pvst
! Primary Root (Priority 24576) for VLAN 30 where HQ-DSW2 is HSRP Active
spanning-tree vlan 30 priority 24576
! Secondary Root (Priority 28672) for VLANs 10, 20, 50, 99, 100, 999
spanning-tree vlan 10,20,50,99,100,999 priority 28672
! --- Loopback & Layer 3 Routed Uplink to HQ-R1 ---
interface Loopback0
 description OSPF_RID
 ip address 172.16.255.3 255.255.255.255
 exit
```

```
interface GigabitEthernet1/0/1
 description L3_ROUTED_UPLINK_TO_HQ-R1_G0/0/2
 no switchport
 ip address 172.16.0.6 255.255.255.252
 ip ospf network point-to-point
 ip ospf cost 20
 no shutdown
 exit
! --- LACP EtherChannel Member Links & Port-Channel Trunks ---
! Po1: Inter-Switch Trunk to HQ-DSW1 (G1/0/23 - 24)
interface GigabitEthernet1/0/23
 description PO1_MEMBER_TO_HQ-DSW1_G1/0/23
 channel-group 1 mode active
 no shutdown
 exit
interface GigabitEthernet1/0/24
 description PO1_MEMBER_TO_HQ-DSW1_G1/0/24
 channel-group 1 mode active
 no shutdown
 exit
! If PT rejects trunk mode ("encapsulation is Auto"), first enter:
!   switchport trunk encapsulation dot1q
interface Port-channel1
 description LACP_TRUNK_TO_HQ-DSW1
 switchport mode trunk
 switchport trunk native vlan 999
 switchport trunk allowed vlan 10,20,30,50,99,100,999
 switchport nonegotiate
 exit
! Po3: Downlink Trunk to HQ-ASW1 (G1/0/2 - 3; speed/duplex matched to FastEthernet)
! Pruned to the VLANs HQ-ASW1 actually carries (10, 20, 99 + native 999)
interface GigabitEthernet1/0/2
 description PO3_MEMBER_TO_HQ-ASW1_F0/23
 speed 100
 duplex full
 channel-group 3 mode active
 no shutdown
 exit
interface GigabitEthernet1/0/3
 description PO3_MEMBER_TO_HQ-ASW1_F0/24
 speed 100
 duplex full
 channel-group 3 mode active
 no shutdown
 exit
interface Port-channel3
 description LACP_TRUNK_TO_HQ-ASW1
 switchport mode trunk
 switchport trunk native vlan 999
 switchport trunk allowed vlan 10,20,99,999
 switchport nonegotiate
 exit
! Po5: Downlink Trunk to HQ-ASW2 (G1/0/4 - 5; speed/duplex matched to FastEthernet)
! Pruned to the VLANs HQ-ASW2 actually carries (30, 50, 99 + native 999)
interface GigabitEthernet1/0/4
 description PO5_MEMBER_TO_HQ-ASW2_F0/23
 speed 100
 duplex full
 channel-group 5 mode active
 no shutdown
 exit
interface GigabitEthernet1/0/5
 description PO5_MEMBER_TO_HQ-ASW2_F0/24
 speed 100
 duplex full
 channel-group 5 mode active
 no shutdown
 exit
interface Port-channel5
```

```
 description LACP_TRUNK_TO_HQ-ASW2
 switchport mode trunk
 switchport trunk native vlan 999
 switchport trunk allowed vlan 30,50,99,999
 switchport nonegotiate
 exit
! --- Park All Unused Physical Ports in BLACKHOLE VLAN 888 & Disable ---
interface range GigabitEthernet1/0/6 - 22, GigabitEthernet1/1/1 - 4
 description UNUSED_BLACKHOLE_PORT
 switchport mode access
 switchport access vlan 888
 shutdown
 exit
! Identical policy applied on HQ-DSW2, which is HSRP Active for VLAN 30
! --- Inter-VLAN Security Extended Named ACL ---
! Restricts HR/Sales (VLAN 30) from the Server Farm (except DHCP/HTTP to HQ-SRV1)
! and blocks all access from VLAN 30 to Campus/Branch Management VLANs
ip access-list extended ACL_RESTRICT_HR
 remark Permit DHCP Discover/Request broadcasts (relayed by ip helper-address)
 permit udp host 0.0.0.0 host 255.255.255.255 eq bootps
 remark Permit DHCP unicast renewals to HQ-SRV1 (client port 68 to server port 67)
 permit udp 10.10.30.0 0.0.0.255 host 10.10.50.10 eq bootps
 remark Permit HTTP Web Access to HQ-SRV1
 permit tcp 10.10.30.0 0.0.0.255 host 10.10.50.10 eq www
 remark Block all other access to Server Farm VLAN 50
 deny ip 10.10.30.0 0.0.0.255 10.10.50.0 0.0.0.255
 remark Block all access to HQ Management VLAN 99 and Branch Management VLAN 199
 deny ip 10.10.30.0 0.0.0.255 10.10.99.0 0.0.0.255
 deny ip 10.10.30.0 0.0.0.255 10.20.199.0 0.0.0.255
 remark Permit all other corporate and internet traffic
 permit ip any any
 exit
! --- Switched Virtual Interfaces (SVIs), HSRP v2, & DHCP Relay ---
! VLAN 10: Engineering Data (HSRP Standby - Default Priority 100)
interface Vlan10
 description ENG_DATA_SVI_HSRP_STANDBY
 ip address 10.10.10.3 255.255.255.0
 ip helper-address 10.10.50.10
 standby version 2
 standby 10 ip 10.10.10.1
 standby 10 priority 100
 no shutdown
 exit
! VLAN 20: VoIP Voice (HSRP Standby - Default Priority 100)
interface Vlan20
 description VOIP_VOICE_SVI_HSRP_STANDBY
 ip address 10.10.20.3 255.255.255.0
 ip helper-address 10.10.50.10
 standby version 2
 standby 20 ip 10.10.20.1
 standby 20 priority 100
 no shutdown
 exit
! VLAN 30: HR & Sales (HSRP Active - Priority 110 with Preemption; ACL Applied Inbound)
interface Vlan30
 description HR_SALES_SVI_HSRP_ACTIVE
 ip address 10.10.30.3 255.255.255.0
 ip helper-address 10.10.50.10
 ip access-group ACL_RESTRICT_HR in
 standby version 2
 standby 30 ip 10.10.30.1
 standby 30 priority 110
 standby 30 preempt
 no shutdown
 exit
! VLAN 50: Server Farm (HSRP Standby - Default Priority 100)
interface Vlan50
 description SERVERS_SVI_HSRP_STANDBY
 ip address 10.10.50.3 255.255.255.0
```

```
 standby version 2
 standby 50 ip 10.10.50.1
 standby 50 priority 100
 no shutdown
 exit
! VLAN 99: Campus Management (HSRP Standby - Default Priority 100)
interface Vlan99
 description MGMT_SVI_HSRP_STANDBY
 ip address 10.10.99.3 255.255.255.0
 standby version 2
 standby 99 ip 10.10.99.1
 standby 99 priority 100
 no shutdown
 exit
! VLAN 100: Dedicated Layer 3 OSPF Transit SVI to HQ-DSW1 over Po1
! OSPF runs as broadcast (DR/BDR election): Packet Tracer does not accept point-to-point on SVIs
interface Vlan100
 description L3_OSPF_TRANSIT_SVI_TO_HQ-DSW1
 ip address 172.16.0.10 255.255.255.252
 no shutdown
 exit
! --- OSPFv2 Area 0 Configuration ---
router ospf 1
 router-id 172.16.255.3
 ! Suppress OSPF Hellos on all end-user/server VLAN SVIs
 passive-interface default
 no passive-interface GigabitEthernet1/0/1
 no passive-interface Vlan100
 ! Advertise Loopback, Core Transit Links, and Campus Subnets into Area 0
 network 172.16.255.3 0.0.0.0 area 0
 network 172.16.0.4 0.0.0.3 area 0
 network 172.16.0.8 0.0.0.3 area 0
 network 10.10.10.0 0.0.0.255 area 0
 network 10.10.20.0 0.0.0.255 area 0
 network 10.10.30.0 0.0.0.255 area 0
 network 10.10.50.0 0.0.0.255 area 0
 network 10.10.99.0 0.0.0.255 area 0
 exit
! --- NTP Client Synchronization ---
ntp server 172.16.255.1
! --- VTY Management Access Control List ---
ip access-list standard ACL_VTY_MGMT
 remark Permit SSH only from HQ and Branch Management VLANs
 permit 10.10.99.0 0.0.0.255
 permit 10.20.199.0 0.0.0.255
 deny any
 exit
! --- Console & VTY Line Hardening ---
line console 0
 logging synchronous
 exec-timeout 5 0
 login local
 exit
line vty 0 15
 access-class ACL_VTY_MGMT in
 logging synchronous
 exec-timeout 5 0
 transport input ssh
 login local
 exit
end
write memory
```

6.6. HQ ACCESS SWITCH 1: HQ-ASW1 (Cisco Catalyst 2960-24TT)

Architectural Role & Key Functions: Layer 2 Access Switch (Engineering Data VLAN 10 & VoIP Voice VLAN 20), LACP Uplinks (Po2 & Po3), Port Security, and NTP Client

```
enable
configure terminal
! --- Baseline Device Hardening ---
hostname HQ-ASW1
no ip domain-lookup
ip domain-name ccna.lab
enable secret CiscoCCNA#Enable1
service password-encryption
username admin privilege 15 secret CiscoCCNA#Admin1
! Banner uses '#' as the delimiter so pasted text does not leave a stray "C"
banner motd #
**************************************************************************
* UNAUTHORIZED ACCESS TO THIS NETWORK DEVICE IS PROHIBITED.              *
* HQ-ASW1 Access Switch 1 - Authorized Personnel Only.                   *
**************************************************************************
#
! Generate 2048-bit RSA keys and enforce SSHv2
! (If PT rejects the one-line form, enter "crypto key generate rsa" and answer 2048)
crypto key generate rsa general-keys modulus 2048
ip ssh version 2
ip ssh time-out 60
ip ssh authentication-retries 3
vtp mode transparent
spanning-tree mode rapid-pvst
! --- VLAN Database Provisioning (only the VLANs this switch carries) ---
vlan 10
 name ENG_DATA
 exit
vlan 20
 name VOIP_VOICE
 exit
vlan 99
 name MGMT
 exit
vlan 888
 name BLACKHOLE
 exit
vlan 999
 name NATIVE_TRUNK
 exit
! --- LACP Uplink Trunks ---
! Po2: Uplink to HQ-DSW1 (F0/21 - 22)
interface FastEthernet0/21
 description PO2_MEMBER_TO_HQ-DSW1_G1/0/2
 speed 100
 duplex full
 channel-group 2 mode active
 no shutdown
 exit
interface FastEthernet0/22
 description PO2_MEMBER_TO_HQ-DSW1_G1/0/3
 speed 100
 duplex full
 channel-group 2 mode active
 no shutdown
 exit
interface Port-channel2
 description LACP_UPLINK_TO_HQ-DSW1
 switchport mode trunk
 switchport trunk native vlan 999
 switchport trunk allowed vlan 10,20,99,999
 switchport nonegotiate
 exit
```

```
! Po3: Uplink to HQ-DSW2 (F0/23 - 24)
interface FastEthernet0/23
 description PO3_MEMBER_TO_HQ-DSW2_G1/0/2
 speed 100
 duplex full
 channel-group 3 mode active
 no shutdown
 exit
interface FastEthernet0/24
 description PO3_MEMBER_TO_HQ-DSW2_G1/0/3
 speed 100
 duplex full
 channel-group 3 mode active
 no shutdown
 exit
interface Port-channel3
 description LACP_UPLINK_TO_HQ-DSW2
 switchport mode trunk
 switchport trunk native vlan 999
 switchport trunk allowed vlan 10,20,99,999
 switchport nonegotiate
 exit
! --- Active Edge Port: IP Phone (VLAN 20) + Engineering PC (VLAN 10) ---
interface FastEthernet0/1
 description ENG_PC1_AND_IP_PHONE_7960
 switchport mode access
 switchport access vlan 10
 switchport voice vlan 20
 ! Port Security allows up to 5 MAC addresses: the PC, the IP Phone (whose MAC can be
 ! learned in both the data and voice VLANs), plus headroom so a phone or PC swap
 ! does not trigger a violation. Restrict mode drops (and logs) frames from extra MACs.
 switchport port-security
 switchport port-security maximum 5
 switchport port-security mac-address sticky
 switchport port-security violation restrict
 ! RSTP Edge Optimization & Protection
 spanning-tree portfast
 spanning-tree bpduguard enable
 no shutdown
 exit
! --- Park All Unused Ports in BLACKHOLE VLAN 888 & Disable ---
interface range FastEthernet0/2 - 20, GigabitEthernet0/1 - 2
 description UNUSED_BLACKHOLE_PORT
 switchport mode access
 switchport access vlan 888
 shutdown
 exit
! --- Management SVI & Default Gateway ---
interface Vlan99
 description MGMT_SVI
 ip address 10.10.99.11 255.255.255.0
 no shutdown
 exit
ip default-gateway 10.10.99.1
! --- NTP Client Synchronization (reached via Management SVI gateway) ---
ntp server 172.16.255.1
! --- VTY Management Access Control List ---
ip access-list standard ACL_VTY_MGMT
 remark Permit SSH only from HQ and Branch Management VLANs
 permit 10.10.99.0 0.0.0.255
 permit 10.20.199.0 0.0.0.255
 deny any
 exit
! --- Console & VTY Line Hardening ---
line console 0
 logging synchronous
 exec-timeout 5 0
 login local
 exit
```

```
line vty 0 15
 access-class ACL_VTY_MGMT in
 logging synchronous
 exec-timeout 5 0
 transport input ssh
 login local
 exit
end
write memory
```

6.7. HQ ACCESS SWITCH 2: HQ-ASW2 (Cisco Catalyst 2960-24TT)

Architectural Role & Key Functions: Layer 2 Access Switch (HR/Sales VLAN 30 & Server Farm VLAN 50), LACP Uplinks (Po4 & Po5), Port Security, and NTP Client

---

```
enable
configure terminal
! --- Baseline Device Hardening ---
hostname HQ-ASW2
no ip domain-lookup
ip domain-name ccna.lab
enable secret CiscoCCNA#Enable1
service password-encryption
username admin privilege 15 secret CiscoCCNA#Admin1
! Banner uses '#' as the delimiter so pasted text does not leave a stray "C"
banner motd #
**************************************************************************
* UNAUTHORIZED ACCESS TO THIS NETWORK DEVICE IS PROHIBITED.              *
* HQ-ASW2 Access Switch 2 - Authorized Personnel Only.                   *
**************************************************************************
#
! Generate 2048-bit RSA keys and enforce SSHv2
! (If PT rejects the one-line form, enter "crypto key generate rsa" and answer 2048)
crypto key generate rsa general-keys modulus 2048
ip ssh version 2
ip ssh time-out 60
ip ssh authentication-retries 3
vtp mode transparent
spanning-tree mode rapid-pvst
! --- VLAN Database Provisioning (only the VLANs this switch carries) ---
vlan 30
 name HR_SALES
 exit
vlan 50
 name SERVERS
 exit
vlan 99
 name MGMT
 exit
vlan 888
 name BLACKHOLE
 exit
vlan 999
 name NATIVE_TRUNK
 exit
! --- LACP Uplink Trunks ---
! Po4: Uplink to HQ-DSW1 (F0/21 - 22)
interface FastEthernet0/21
 description PO4_MEMBER_TO_HQ-DSW1_G1/0/4
 speed 100
 duplex full
 channel-group 4 mode active
 no shutdown
 exit
interface FastEthernet0/22
 description PO4_MEMBER_TO_HQ-DSW1_G1/0/5
 speed 100
 duplex full
 channel-group 4 mode active
 no shutdown
 exit
interface Port-channel4
 description LACP_UPLINK_TO_HQ-DSW1
 switchport mode trunk
 switchport trunk native vlan 999
 switchport trunk allowed vlan 30,50,99,999
 switchport nonegotiate
 exit
```

```
! Po5: Uplink to HQ-DSW2 (F0/23 - 24)
interface FastEthernet0/23
 description PO5_MEMBER_TO_HQ-DSW2_G1/0/4
 speed 100
 duplex full
 channel-group 5 mode active
 no shutdown
 exit
interface FastEthernet0/24
 description PO5_MEMBER_TO_HQ-DSW2_G1/0/5
 speed 100
 duplex full
 channel-group 5 mode active
 no shutdown
 exit
interface Port-channel5
 description LACP_UPLINK_TO_HQ-DSW2
 switchport mode trunk
 switchport trunk native vlan 999
 switchport trunk allowed vlan 30,50,99,999
 switchport nonegotiate
 exit
! --- Active Edge Port: HR/Sales Workstation (HQ-PC2 in VLAN 30) ---
interface FastEthernet0/1
 description HR_SALES_PC2_PORT
 switchport mode access
 switchport access vlan 30
 ! Strict Port Security: Single Sticky MAC, Shutdown (Err-Disable) on violation
 switchport port-security
 switchport port-security maximum 1
 switchport port-security mac-address sticky
 switchport port-security violation shutdown
 spanning-tree portfast
 spanning-tree bpduguard enable
 no shutdown
 exit
! --- Active Server Port: HQ-SRV1 (VLAN 50) ---
interface FastEthernet0/10
 description HQ-SRV1_CENTRAL_DHCP_AND_WEB_SERVER
 switchport mode access
 switchport access vlan 50
 spanning-tree portfast
 spanning-tree bpduguard enable
 no shutdown
 exit
! --- Park All Unused Ports in BLACKHOLE VLAN 888 & Disable ---
interface range FastEthernet0/2 - 9, FastEthernet0/11 - 20, GigabitEthernet0/1 - 2
 description UNUSED_BLACKHOLE_PORT
 switchport mode access
 switchport access vlan 888
 shutdown
 exit
! --- Management SVI & Default Gateway ---
interface Vlan99
 description MGMT_SVI
 ip address 10.10.99.12 255.255.255.0
 no shutdown
 exit
ip default-gateway 10.10.99.1
! --- NTP Client Synchronization (reached via Management SVI gateway) ---
ntp server 172.16.255.1
! --- VTY Management Access Control List ---
ip access-list standard ACL_VTY_MGMT
 remark Permit SSH only from HQ and Branch Management VLANs
 permit 10.10.99.0 0.0.0.255
 permit 10.20.199.0 0.0.0.255
 deny any
 exit
! --- Console & VTY Line Hardening ---
```

```
line console 0
 logging synchronous
 exec-timeout 5 0
 login local
 exit
line vty 0 15
 access-class ACL_VTY_MGMT in
 logging synchronous
 exec-timeout 5 0
 transport input ssh
 login local
 exit
end
write memory
```

6.8. BRANCH ACCESS SWITCH 1: BR-ASW1 (Cisco Catalyst 2960-24TT)

Architectural Role & Key Functions: Branch Layer 2 Access Switch (VLAN 110 Data, VLAN 199 Mgmt), 802.1Q Trunk to BR-R1 (Router-on-a-Stick), Rapid PVST+ Root, Port Security, and NTP Client

---

```
enable
configure terminal
! --- Baseline Device Hardening ---
hostname BR-ASW1
no ip domain-lookup
ip domain-name ccna.lab
enable secret CiscoCCNA#Enable1
service password-encryption
username admin privilege 15 secret CiscoCCNA#Admin1
! Banner uses '#' as the delimiter so pasted text does not leave a stray "C"
banner motd #
**************************************************************************
* UNAUTHORIZED ACCESS TO THIS NETWORK DEVICE IS PROHIBITED.              *
* BR-ASW1 Branch Access Switch - Authorized Personnel Only.              *
**************************************************************************
#
! Generate 2048-bit RSA keys and enforce SSHv2
! (If PT rejects the one-line form, enter "crypto key generate rsa" and answer 2048)
crypto key generate rsa general-keys modulus 2048
ip ssh version 2
ip ssh time-out 60
ip ssh authentication-retries 3
vtp mode transparent
spanning-tree mode rapid-pvst
spanning-tree vlan 110,199,999 priority 24576
! --- VLAN Database Provisioning ---
vlan 110
 name BR_DATA
 exit
vlan 199
 name BR_MGMT
 exit
vlan 888
 name BLACKHOLE
 exit
vlan 999
 name NATIVE_TRUNK
 exit
! --- 802.1Q Uplink Trunk to BR-R1 (Router-on-a-Stick) ---
interface GigabitEthernet0/1
 description 8021Q_TRUNK_TO_BR-R1_G0/1
 switchport mode trunk
 switchport trunk native vlan 999
 switchport trunk allowed vlan 110,199,999
 switchport nonegotiate
 no shutdown
 exit
! --- Active Edge Port: Branch Workstation (BR-PC1 in VLAN 110) ---
interface FastEthernet0/1
 description BRANCH_PC1_PORT
 switchport mode access
 switchport access vlan 110
 switchport port-security
 switchport port-security maximum 1
 switchport port-security mac-address sticky
 switchport port-security violation shutdown
 spanning-tree portfast
 spanning-tree bpduguard enable
 no shutdown
 exit
! --- Park All Unused Ports in BLACKHOLE VLAN 888 & Disable ---
interface range FastEthernet0/2 - 24, GigabitEthernet0/2
 description UNUSED_BLACKHOLE_PORT
```

```
 switchport mode access
 switchport access vlan 888
 shutdown
 exit
! --- Branch Management SVI & Default Gateway ---
interface Vlan199
 description BRANCH_MGMT_SVI
 ip address 10.20.199.11 255.255.255.0
 no shutdown
 exit
ip default-gateway 10.20.199.1
! --- NTP Client Synchronization (reached via Management SVI gateway) ---
ntp server 172.16.255.1
! --- VTY Management Access Control List ---
ip access-list standard ACL_VTY_MGMT
 remark Permit SSH only from HQ and Branch Management VLANs
 permit 10.10.99.0 0.0.0.255
 permit 10.20.199.0 0.0.0.255
 deny any
 exit
! --- Console & VTY Line Hardening ---
line console 0
 logging synchronous
 exec-timeout 5 0
 login local
 exit
line vty 0 15
 access-class ACL_VTY_MGMT in
 logging synchronous
 exec-timeout 5 0
 transport input ssh
 login local
 exit
end
write memory
```

6.9. SERVER & ENDPOINT SETTINGS (Packet Tracer GUI)

Architectural Role & Key Functions: Desktop and Services settings for HQ-SRV1, ISP-WEB and the endpoints. Phase 3 depends on these; the IOS blocks above do not configure them.

HQ-SRV1 — Desktop › IP Configuration (Static)

IPv4 Address	Subnet Mask	Default Gateway	DNS Server
10.10.50.10	255.255.255.0	10.10.50.1 (HSRP VIP — required for replies to the DHCP relays and NAT traffic)	203.0.113.200

HQ-SRV1 — Services › DHCP (Service: On)

Pool Name	Default Gateway	DNS Server	Start IP Address	Subnet Mask	Maximum Users	TFTP Server
ENG_DATA_POOL	10.10.10.1	203.0.113.200	10.10.10.50	255.255.255.0	151	0.0.0.0
VOICE_POOL	10.10.20.1	203.0.113.200	10.10.20.50	255.255.255.0	151	10.10.50.10
HR_SALES_POOL	10.10.30.1	203.0.113.200	10.10.30.50	255.255.255.0	151	0.0.0.0

151 users from `.50` gives the `.50`–`.200` range in Section 3. Requests arrive relayed, so each is matched to the pool whose subnet contains the relay address (`10.10.x.2` from HQ-DSW1 or `10.10.x.3` from HQ-DSW2). Leave the built-in `serverPool` as it is. Services › HTTP: On.

ISP-WEB — Desktop › IP Configuration (Static) & Services

Setting	Value
IPv4 Address / Mask	203.0.113.200 / 255.255.255.128
Default Gateway	203.0.113.129
DNS Server	203.0.113.200
Services › DNS (On) — A Records	web.isp.lab → 203.0.113.200 hqweb.ccna.lab → 203.0.113.66 (HQ-SRV1 via Static NAT)
Services › HTTP	On

Endpoints

Device	Setting	Expected Result
HQ-PC1, HQ-PC2, BR-PC1	Desktop › IP Configuration › DHCP	Lease from `.50` upward in VLAN 10, 30 and 110 respectively
HQ-PH1	Power adapter installed (see Section 1)	Lease in VLAN 20 via Voice VLAN

Static NAT test: from ISP-WEB, open `http://203.0.113.66` in the Web Browser — the HQ-SRV1 page should load and `show ip nat translations` on HQ-R1 shows the hit. Internal hosts should browse `http://10.10.50.10` directly; NAT hairpinning to the public address is not supported.

Appendix A: Test & Verification

Run these checks after the build is complete, in order. Each stage depends on the one before it, so the first failing check points to the layer with the problem. **Run on** is the device whose CLI (or Desktop › Command Prompt / Web Browser for PCs and servers) you use.

> **Before you start:** Click **Fast Forward Time** a few times so STP, LACP, OSPF and NTP converge. The first ping to a new destination often loses one packet while ARP resolves; that is normal. DHCP leases may start above `.50` if earlier leases are still held by the server.

A.1 Layer 2: VLANs, Trunks, EtherChannel, STP & HSRP

Run on	Command / Action	Expected Result
All switches	`show vtp status`	`VTP Operating Mode : Transparent`
HQ-ASW1	`show vlan brief`	VLANs 10, 20, 99, 888, 999 only; `Fa0/1` in VLAN 10; `Fa0/2–20`, `Gi0/1–2` in VLAN 888
HQ-ASW2	`show vlan brief`	VLANs 30, 50, 99, 888, 999 only; `Fa0/1` in VLAN 30, `Fa0/10` in VLAN 50
HQ-DSW1	`show etherchannel summary`	`Po1(SU)`, `Po2(SU)`, `Po4(SU)`, Protocol `LACP`, every member port flagged `(P)`
HQ-DSW2	`show etherchannel summary`	`Po1(SU)`, `Po3(SU)`, `Po5(SU)`, every member `(P)`
HQ-ASW1 / HQ-ASW2	`show etherchannel summary`	`Po2(SU)` + `Po3(SU)` on HQ-ASW1; `Po4(SU)` + `Po5(SU)` on HQ-ASW2
HQ-DSW1	`show interfaces trunk`	All Port-channels `trunking`, `802.1q`, native VLAN `999`; allowed: `Po1` `10,20,30,50,99-100,999`, `Po2` `10,20,99,999`, `Po4` `30,50,99,999`
BR-ASW1	`show interfaces trunk`	`Gi0/1` trunking, native `999`, allowed `110,199,999`
HQ-DSW1	`show spanning-tree vlan 10`	`This bridge is the root` (repeat for VLANs 20, 50, 99)
HQ-DSW2	`show spanning-tree vlan 30`	`This bridge is the root`
BR-ASW1	`show spanning-tree vlan 110`	`This bridge is the root`
HQ-DSW1	`show standby brief`	`Vl10`, `Vl20`, `Vl50`, `Vl99`: priority `110`, `P`, `Active`; `Vl30`: priority `100`, `Standby`
HQ-DSW2	`show standby brief`	`Vl30`: priority `110`, `P`, `Active`; all other groups `Standby`

A.2 Layer 3: OSPF & Static Routing

Run on	Command / Action	Expected Result
HQ-R1	`show ip ospf neighbor`	Three neighbors: `172.16.255.2` `FULL/ -` on `Gi0/0/1`; `172.16.255.3` `FULL/ -` on `Gi0/0/2`; `172.16.255.4` `FULL/DR` or `FULL/BDR` on `Vlan1`
HQ-DSW1	`show ip ospf neighbor`	`172.16.255.1` `FULL/ -` on `Gi1/0/1`; `172.16.255.3` `FULL/DR` or `FULL/BDR` on `Vlan100`
BR-R1	`show ip ospf neighbor`	`172.16.255.1` `FULL/DR` or `FULL/BDR` on `Gi0/0`

Run on	Command / Action	Expected Result
HQ-R1	`show ip protocols`	`Routing Protocol is "ospf 1"`, Router ID `172.16.255.1`, `It is an area border and autonomous system boundary router`
HQ-R1	`show ip route`	`S* 0.0.0.0/0 [1/0] via 203.0.113.1`; `O 10.10.x.0/24 [110/11] via 172.16.0.2`; `O 10.20.110.0/24 [110/2] via 172.16.1.2, Vlan1`
HQ-DSW1	`show ip route`	`O*E2 0.0.0.0/0`; `O IA 10.20.110.0/24` and `10.20.199.0/24`
BR-R1	`show ip route`	`O*E2 0.0.0.0/0 via 172.16.1.1`; `O IA 10.10.10.0/24` (and the other HQ subnets)
ISP-R1	`show ip route`	`S 203.0.113.64/29 [1/0] via 203.0.113.2`; `Gateway of last resort is not set` (correct for the ISP)

A.3 Services: DHCP, DNS, NTP & NAT

Run on	Command / Action	Expected Result
HQ-PC1	`ipconfig /all` (IP Configuration set to DHCP)	IPv4 `10.10.10.50` or higher, gateway `10.10.10.1`, DHCP server `10.10.50.10`, DNS `203.0.113.200`
HQ-PC2	`ipconfig /all`	IPv4 `10.10.30.50` or higher, gateway `10.10.30.1`, DHCP server `10.10.50.10`
BR-PC1	`ipconfig /all`	IPv4 `10.20.110.50` or higher, gateway `10.20.110.1`, DNS `203.0.113.200`
BR-R1	`show ip dhcp binding`	BR-PC1's MAC bound to its `10.20.110.x` lease
HQ-PC1	`nslookup web.isp.lab`	`Name: web.isp.lab`, `Address: 203.0.113.200`
HQ-R1	`show ntp status`	`Clock is synchronized, stratum 2` (NTP master)
HQ-DSW1, BR-R1, HQ-ASW1	`show ntp status`	`Clock is synchronized, stratum 3, reference is 172.16.255.1` (fast-forward if not yet synchronized)
HQ-PC1	`ping 203.0.113.200`	4 replies, `TTL=125`
HQ-R1	`show ip nat translations`	`icmp 203.0.113.65:x 10.10.10.x:x 203.0.113.200:x` entries plus the permanent static line `--- 203.0.113.66 10.10.50.10 ---`
BR-PC1	`ping 203.0.113.200`	4 replies; HQ-R1 then shows a translation with inside local `10.20.110.x`
HQ-PC1	Web Browser › `http://web.isp.lab`	ISP-WEB's page loads (proves DNS + PAT end to end)
ISP-WEB	Web Browser › `http://203.0.113.66`	HQ-SRV1's page loads; HQ-R1 `show ip nat translations` shows `tcp 203.0.113.66:80 10.10.50.10:80`

A.4 Security: ACLs, Port Security & SSH

Inter-VLAN ACL — run from HQ-PC2 (VLAN 30)

Run on	Command / Action	Expected Result	Why
HQ-PC2	`ping 10.10.50.10`	**Fails**	ICMP to the Server Farm is denied
HQ-PC2	Web Browser › `http://10.10.50.10`	**Loads**	HTTP to HQ-SRV1 is permitted
HQ-PC2	`ping 10.10.99.2`	**Fails**	HQ Management VLAN is denied
HQ-PC2	`ping 203.0.113.200`	Replies	Internet traffic is permitted
HQ-DSW2	`show access-lists ACL_RESTRICT_HR`	Match counters on the `www` permit and the `deny` lines increase	HQ-DSW2 is the active VLAN 30 gateway

Run on	Command / Action	Expected Result
HQ-ASW1	`show port-security interface fa0/1`	`Port Status : Secure-up`, `Violation Mode : Restrict`, `Maximum MAC Addresses : 5`
HQ-ASW2 / BR-ASW1	`show port-security interface fa0/1`	`Secure-up`, `Violation Mode : Shutdown`, `Maximum MAC Addresses : 1`, `Sticky MAC Addresses : 1`
HQ-ASW1	`show port-security address`	HQ-PC1's MAC (and the IP Phone's) listed as `SecureSticky` on `Fa0/1`
Any router or switch	`show ip ssh`	`SSH Enabled - version 2.0`
Any router or switch	`show access-lists ACL_VTY_MGMT`	Permits `10.10.99.0 0.0.0.255` and `10.20.199.0 0.0.0.255`, then `deny any`
HQ-ASW2 (optional)	Swap HQ-PC2 for a new PC on `Fa0/1` and ping	`Fa0/1` goes `err-disabled`; recover by reconnecting HQ-PC2, then `shutdown` / `no shutdown` on the port
BR-ASW1 (optional)	Connect a spare switch to `Fa0/1`	BPDU Guard puts `Fa0/1` into `err-disabled`; recover with `shutdown` / `no shutdown`

A.5 Redundancy & Failover

Start a continuous ping from HQ-PC1 (`ping -t 203.0.113.200`, or repeat `ping 203.0.113.200`) and run each test while it is going. Restore each failure before starting the next. When every check in this appendix matches, the lab is working end to end.

Break (Run on)	What to Check	Restore
HQ-R1: `interface g0/0/0` › `shutdown`	HQ-R1 `show ip route` shows `S* 0.0.0.0/0 [5/0] via 198.51.100.1`; ISP-R1 shows `203.0.113.64/29 [5/0] via 198.51.100.2`; the ping resumes after a few drops; `show ip nat translations` still uses `203.0.113.65`	`no shutdown` — routes return to `[1/0]`
HQ-DSW1: `interface vlan 10` › `shutdown`	HQ-DSW2 `show standby brief` shows `Vl10 Active`; the ping continues	`no shutdown` — HQ-DSW1 preempts back to Active

Break (Run on)	What to Check	Restore
HQ-DSW1: `interface g1/0/2` › `shutdown`	`show etherchannel summary`: `Po2(SU)` stays up with `Gi1/0/3(P)` and `Gi1/0/2(D)`; the ping continues	`no shutdown`
HQ-DSW1: `interface g1/0/1` › `shutdown`	HQ-R1 `show ip route ospf` reaches `10.10.x.0/24` via `172.16.0.6` (cost 21); the ping continues	`no shutdown` — routes return via `172.16.0.2`